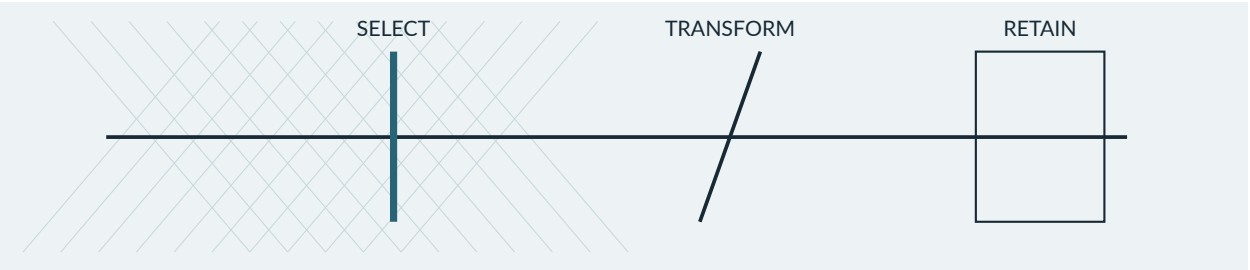
TOM KLOOTWIJK / FORMAL ARCHITECTURE

The Klootwijk Architectural Paradigm

Two preparaats. Two clocks.
One history-bearing architecture.

An axiomatic specification of optical selection,
state conversion, crosshatch mechanics,
and hourglass recovery.



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Conceptual source: the author-supplied dialogue *trewel.pdf*.

Architectural statement

The Klootwijk paradigm organizes an event as the encounter between a structured latent capacity and an activating process. Geometry selects and redirects the encounter; coupled material states express it; a fast clock executes it; a slow clock retains its history. Recovery is determined by what passed through the architecture and how long the active configuration was maintained.

The architecture is specified here through definitions, constitutive postulates, and consequences derived from those postulates. Its recurring names—Preparaat 1, Preparaat 2, OTAN2, the focal cross, the walnut crosshatch, the hourglass, the photographic carbon copy, and Clock Alpha and Clock Beta—are assigned explicit mathematical roles. The original prompts establish this sequence from log-polar encoding through optical selection to material interaction and dual-clock self-reference. [S, pp. 3, 6, 14, 16, 22, 26–28]

The central proposition

Geometry, activation, and retained history form one coupled architecture. Geometry determines available paths. The interaction of the two preparaat roles changes local state. That change can reorient an anisotropic network, thereby changing transport. Transport integrates the duration of activity, and the resulting memory sets the recovery trajectory. A representation channel indexes the same event without conflating spatial coordinates, material quantities, and stored records.

Six architectural commitments

A1. **Paired roles:** a latent structural capacity and an activating or mobilized process must be identified in each module.

A2. **Boundary mediation:** an event enters through specified interfaces rather than acting everywhere without a path.

A3. **Coupled execution:** optical, material, mechanical, and memory variables update through declared interfaces.

A4. **Orthogonal configuration:** the crosshatch admits two designated orientations separated by a quarter-turn.

A5. **Dual temporal scales:** active response and retained history evolve on distinct time scales.

A6. **History-dependent return:** recovery depends on the accumulated state, not on elapsed time alone.

Reading map

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1 Vocabulary and semantic roles

Source term	Meaning in the specification
Preparaat 1	Structured capacity: the latent, supporting, or retaining role, written P_1.
Preparaat 2	Activating or mobilized role, written P_2, whose interaction with P_1 drives a state transition.
Two-sided lane	A rectangular domain with two designated opposing exchange interfaces, Γ_0 and Γ_H.
Focal cross / double Hadamard intersection	Co-location of indexed fields; when represented as arrays, two successive componentwise products.
Inuit vertical slit	The source's name for the oriented aperture operator $\mathcal{S}_{w,\gamma}$. Vertical is a specified orientation of that operator.
Tilt-shift	Optical plane orientation and image-plane displacement, represented by $\mathcal{O}_{\phi,d}$.
OTAN2	The architecture's regularized logarithmic-radius, quadrant-phase, and angular-validity encoder.
Walnut L-system	A generative description interpreted as a material crosshatch graph; dynamics take place on the resulting graph.
Chemical atomic clock	A pair consisting of a reference tick count $N(t)$ and an accumulated chemical progress variable $\vartheta(t)$.
Clock Alpha / Clock Beta	Fast active state a and slow retained state b, respectively.
Hourglass tilt	Orientation-dependent transport with accumulated dwell duration X and memory occupancy m.
Vantablack / Vanta-state	The high-optical-depth regime of the architecture's absorption model, $\mathcal{T} \gg 1$.
Flash-frozen carbon copy / petrification	A time-indexed record whose stored value is held fixed after commitment.
White Dove / stand by the signal	The record-preserving projection Ψ_{dove} defined in Section 12.
Trewel / LNR	Source identifiers retained without introducing an unprovided acronym expansion.

Role, constituent, and conversion state

The preparaat names are *architectural roles*. A material realization separately identifies its scaffold, mobile constituent, and conversion variable. For a chosen convertible population, the state convention is

$$P_1 = P_\star(1 - c), \qquad P_2 = P_\star c, \qquad 0 \le c \le 1. \tag{1.1}$$

Here $P_\star$ is the total amount density of that population. In the lipid–carbohydrate realization, the scaffold and mobile phase have their own material identities; c describes a specified transition within that realization. This typing allows the same architecture to be instantiated in a material lattice, an imaging module, or a computational record without assigning one substance to every role.

Source correspondence: *trewel.pdf*, pp. 6–18, 22–29..

2 Typed state and composition

State space

Let the continuum state and its information registers be collected as

$$\mathsf{x}(t) = \big[T, c, \mathbf{u}, p, q, m, a, b, D, \mathscr{H}\big](t). \tag{2.1}$$

$\mathbf{u}$ is displacement, p is pore pressure, $q \in \{0, 1\}$ is orientation, D is a committed record, and $\mathscr{H}$ is an ordered history. The irradiances I_+ and I_- and the stresses $\boldsymbol{\sigma}$ are coupled fields obtained from the optical and constitutive equations. Spatial encoding returns a separate coordinate record ξ.

Symbol	Quantity	Unit
$x, y, z, H, r_0, \epsilon, w$	Coordinates and geometric scales	m
$t, X, \tau_\alpha, \tau_\beta, \tau_r$	Durations and time constants	s
T; c, m, q	Temperature; dimensionless states	K; 1
$I_\pm, I_\star$	Irradiance and reference irradiance	W m^{-2}
μ_a; k_f, k_b	Absorption; kinetic rates	m^{-1}; s^{-1}
$P_\star$	Convertible amount density	mol m^{-3}
$\boldsymbol{\sigma}, p, p_c$	Stress and pressures	Pa
$\mathbf{K}$; η_f; $\mathbf{j}$	Permeability; viscosity; Darcy flux	m^2; Pa s; m s^{-1}
ρ_f; $M_{\mathrm{tr}}, M_\star$	Fluid density; masses	kg m^{-3}; kg
$\rho_\epsilon, \theta, \vartheta$	Log radius, angle, chemical progress	1, rad, 1

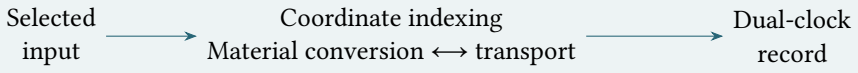
Module interfaces

A module is a tuple

$$\mathscr{M} = (\mathscr{U}, \mathscr{X}, \mathscr{Y}, F, G), \qquad \dot{x} = F(x, u), \quad y = G(x, u). \tag{2.2}$$

Composition connects an output to a compatible input. A coordinate transform therefore acts on coordinate arrays, an absorption law acts on irradiance, and a rotation acts on vectors or tensors. Their synchronization is a product-state update rather than the addition of incompatible quantities.

Master coupling



Dimensionless logarithms

Every logarithm acts on a dimensionless ratio. Examples are $\ln(r/r_0)$, $\ln(1+\beta_t X)$ with $[\beta_t] = \mathrm{s}^{-1}$, and $\ln(1+M/M_\star)$. Spatial regularization ϵ, pore compressibility β_p, and temporal braking β_t are distinct parameters.

3 The two-sided optical lane

Domain and interfaces

Define the lane by

$$\Omega = [0, L_x] \times [0, L_y] \times [0, H], \qquad \Gamma_0 = \{z = 0\}, \quad \Gamma_H = \{z = H\}. \tag{3.1}$$

The two opposing interfaces carry independently specified optical or thermal inputs. Lateral faces have their own boundary conditions. The two-sided searing motif is thus represented by two boundary-driven conversion fronts within one rectangular field.

Oriented slit operator

For transverse position $\mathbf{r} = (x, y)$, let

$$\begin{aligned} \mathbf{n}_\gamma &= (\cos\gamma, \sin\gamma), \quad \mathbf{t}_\gamma = (-\sin\gamma, \cos\gamma), \\ A_{w,\ell,\gamma}(\mathbf{r}) &= \operatorname{rect}\left(\frac{\mathbf{n}_\gamma \cdot (\mathbf{r} - \mathbf{r}_c)}{w}\right) \operatorname{rect}\left(\frac{\mathbf{t}_\gamma \cdot (\mathbf{r} - \mathbf{r}_c)}{\ell}\right). \end{aligned} \tag{3.2}$$

Here $\operatorname{rect}(s) = 1$ for $|s| \le 1/2$ and 0 otherwise. A vertical slit has $\gamma = 0$ and $\ell \gg w$. For an optical field L, define $\mathcal{S}L = A_{w,\ell,\gamma}L$. The normalized transverse kernel $w^{-1}\operatorname{rect}(x/w)$ has a delta-function limit; the finite aperture itself retains its finite width and throughput.

Tilt-shift propagation

The selected field is propagated by a calibrated optical operator,

$$E(\mathbf{r}, t) = \mathcal{O}_{\phi,d}\, \mathcal{S}_{w,\gamma} L(t). \tag{3.3}$$

ϕ specifies plane tilt and d specifies shift. In a planar geometric representation, corresponding image coordinates satisfy

$$\tilde{\mathbf{r}}' \sim \mathbf{H}_{\phi,d}\tilde{\mathbf{r}}, \qquad \tilde{\mathbf{r}} = (x, y, 1)^{\mathsf{T}}, \tag{3.4}$$

with a nonsingular homography on its usable coordinate chart. Focus geometry is specified separately: in the Scheimpflug configuration, the extended object, lens, and image planes share a common line. [2]

The shared focal cross

At each observation location, optical exposure and coordinate encoding are joined as

$$\mathsf{v}(\mathbf{r}, t) = \big(E(\mathbf{r}, t), \operatorname{OTAN2}_{\epsilon,M}(y, x), t\big). \tag{3.5}$$

This common index is the bridge between the material lane and the digital lane. Slit width and foveal scale may be coupled by a design relation $\epsilon = \kappa_w w$ with dimensionless $\kappa_w > 0$.

Source correspondence: *trewel.pdf*, pp. 6–7, 14–16..

4 OTAN2: the coordinate specification

OTAN2 denotes the complete architectural encoder, rather than an alternative name for its angular subroutine. For $\epsilon, r_0, M > 0$, define

$$\boxed{\mathrm{OTAN2}_{\epsilon,M}(y,x) = (\rho_\epsilon, \theta, v_\theta)}, \qquad \rho_\epsilon = \frac{M}{2}\ln\!\left(\frac{x^2+y^2+\epsilon^2}{r_0^2}\right). \tag{4.1}$$

The angular-validity bit is $v_\theta = 1$ when $(x,y) \neq (0,0)$ and 0 otherwise. At the origin, set the stored angle to 0 by convention. The logarithmic radius is finite there:

$$\rho_\epsilon(0,0) = M\ln(\epsilon/r_0). \tag{4.2}$$

The validity bit distinguishes an angle convention from a measured direction.

Octant reduction and quadrant reconstruction

For $(x,y) \neq (0,0)$, put

$$z = \frac{\min(|x|,|y|)}{\max(|x|,|y|)}, \qquad a = \arctan z. \tag{4.3}$$

Reconstruct the phase through conditional selections:

$$\begin{aligned} b &= \begin{cases} \pi/2 - a, & |y| > |x|, \\ a, & |y| \le |x|, \end{cases} \qquad c_\theta = \begin{cases} \pi - b, & x < 0, \\ b, & x \ge 0, \end{cases} \\ \theta &= \begin{cases} -c_\theta, & y < 0, \\ c_\theta, & y \ge 0. \end{cases} \end{aligned} \tag{4.4}$$

This fixes all quadrants and axes, with $\theta \in (-\pi, \pi]$ under the ordinary zero convention. Selection operations can be implemented with masks; whether the compiled kernel has branches is an implementation property.

Zero-based radius and inversion

A foveal-origin coordinate is

$$\bar\rho = \rho_\epsilon - M\ln(\epsilon/r_0) = \frac{M}{2}\ln\!\left(1 + \frac{r^2}{\epsilon^2}\right), \qquad r = \sqrt{x^2+y^2}. \tag{4.5}$$

Its inverse, for $\bar\rho \ge 0$, is

$$r = \epsilon\sqrt{\exp(2\bar\rho/M) - 1}, \qquad (x,y) = (r\cos\theta, r\sin\theta). \tag{4.6}$$

Coordinate consequence. A quarter-turn leaves ρ_ϵ unchanged and advances θ by $\pi/2$ modulo 2π. Far from the regularization scale, $\rho_\epsilon = M\ln(r/r_0) + O(\epsilon^2/r^2)$.

Source correspondence: *trewel.pdf*, pp. 3–6, 12–14..

5 Direct log encoding and DSP realization

Signed logarithmic inputs

Represent nonzero coordinates by signs s_x, s_y and dimensionless log magnitudes

$$\ell_x = \ln(|x|/r_0), \qquad \ell_y = \ln(|y|/r_0), \qquad \ell_\epsilon = \ln(\epsilon/r_0). \tag{5.1}$$

Zero coordinates use log magnitude $-\infty$. The radius can be calculated directly in this representation:

$$\rho_\epsilon = \frac{M}{2}\,\mathrm{LSE}(2\ell_x, 2\ell_y, 2\ell_\epsilon), \tag{5.2}$$

where

$$\mathrm{LSE}(v_1, \dots, v_k) = v_{\max} + \ln \sum_j \exp(v_j - v_{\max}). \tag{5.3}$$

The angular octant ratio is likewise obtained from the log difference,

$$z = \exp\!\left(-|\ell_x - \ell_y|\right), \tag{5.4}$$

with the both-zero case handled by the validity convention. Thus OTAN2 accepts log-encoded polar construction without reconstructing large Cartesian magnitudes.

Polynomial angular engine

For a multiply-accumulate implementation, replace $\arctan z$ by

$$p(z) = z(a_1 + a_3 z^2 + a_5 z^4), \qquad 0 \le z \le 1. \tag{5.5}$$

The coefficient set belongs to the implementation and carries a specified error bound $\sup_{[0,1]} |p(z) - \arctan z| \le \delta_\theta$. Quadrant reconstruction only changes signs and adds constants, so it preserves that angular error bound away from representation wrap. Floating-point and quantization errors are accounted for separately. The reference implementation in Appendix A uses the exact library arctangent of the reduced ratio.

OpenCV interface

Use the inverse equations to build sampling maps, then apply a declared interpolation and border rule. OpenCV documents both geometric remapping and the standard semi-logarithmic `warpPolar` transform. [1] A custom map implements the particular regularization in Equation (4.1); the mathematical transform, sampling lattice, and interpolation rule are three distinct parts of the implementation.

Digital indexing

For $r \le r_{\max}$, a normalized output index can be chosen as

$$j_\rho = (N_\rho - 1)\frac{\bar\rho}{\bar\rho_{\max}}, \qquad j_\theta = N_\theta \frac{\theta \bmod 2\pi}{2\pi}. \tag{5.6}$$

Angular indexing is periodic. The center retains its validity bit through resampling and readout.

6 Two-preparaat conversion and absorption

Conserved conversion population

For the state convention in Section 1, adopt

$$\partial_t c = k_f(T)\, i\,(1-c) - k_b(T) c, \qquad i = \frac{I_+ + I_-}{I_\star}. \tag{6.1}$$

The corresponding amount balances are

$$\partial_t P_1 = -P_\star \partial_t c, \qquad \partial_t P_2 = +P_\star \partial_t c, \qquad P_1 + P_2 = P_\star. \tag{6.2}$$

Input irradiance is a driving field; the conservation statement concerns the designated convertible population. The absorbing two-state transfer law is a local material module within the larger preparaat architecture.

Two-sided attenuation

Use a quasi-static, absorption-only lane with

$$\mu_a(c) = \mu_1(1-c) + \mu_2 c, \qquad \mu_1, \mu_2 \geq 0, \tag{6.3}$$

$$\partial_z I_+ = -\mu_a(c) I_+, \qquad \partial_z I_- = +\mu_a(c) I_-. \tag{6.4}$$

The boundary values are $I_+(0,t) = I_0(t)$ and $I_-(H,t) = I_H(t)$. Integration gives

$$\begin{aligned} I_+(z,t) &= I_0(t) \exp\!\left[-\int_0^z \mu_a(c(s,t))\, \mathrm{d}s\right], \\ I_-(z,t) &= I_H(t) \exp\!\left[-\int_z^H \mu_a(c(s,t))\, \mathrm{d}s\right]. \end{aligned} \tag{6.5}$$

These are Beer–Lambert-type transport equations in the declared absorption model. [3] The source's Vantastate is represented by large optical depth $\mathscr{T} = \int \mu_a \, \mathrm{d}z$, giving a small transmitted fraction $e^{-\mathscr{T}}$.

Double Hadamard intersection

On a common spatial grid, let $\mathbf{f}_1 = \mathbf{1} - \mathbf{c}$, $\mathbf{i}$ be normalized irradiance, and $\mathbf{g}$ a local activation mask. The source's double intersection becomes

$$\mathbf{h} = (\mathbf{f}_1 \odot \mathbf{i}) \odot \mathbf{g}, \qquad \dot{\mathbf{c}} = \mathbf{k}_f \odot \mathbf{h} - \mathbf{k}_b \odot \mathbf{c}. \tag{6.6}$$

Both products are componentwise. The mask and the array index make the location of the encounter explicit.

Conservation and admissible state. At $c = 0$ the rate is nonnegative, and at $c = 1$ it is nonpositive. For nonnegative rates and irradiance, $[0,1]$ is invariant and the two populations retain their prescribed total.

7 Temperature, chemical time, and the front

Thermal evolution and rate law

A thermal realization uses volumetric enthalpy h and conductivity κ:

$$\partial_t h(T,\zeta) = \nabla\cdot(\kappa\nabla T) + Q_{\text{heat}}, \qquad h = \rho_m c_p(T - T_\star) + \rho_m L\zeta. \tag{7.1}$$

The phase fraction ζ has a specified constitutive relation to temperature or its own kinetic law. For stored chemical energy density $\Delta h\, P_\star c$, an energy-consistent closure is

$$Q_{\text{heat}} = \mu_a(I_+ + I_-) - \Delta h\, P_\star \partial_t c + Q_{\text{ext}}, \tag{7.2}$$

where Δh is energy per mole and Q_{ext} is an external volumetric heat source. A thermal boundary is $-\kappa\nabla T\cdot \mathbf{n} = h_c(T - T_{\text{env}}) - q_{\text{in}}$.

For the chemical rate adopt the Arrhenius form [4]

$$k_f(T) = A_f \exp\!\left(-\frac{E_a}{RT}\right), \qquad T > 0. \tag{7.3}$$

For constant $A_f, E_a > 0$, $\mathrm{d}\ln k_f/\mathrm{d}T = E_a/(RT^2) > 0$. T is physical temperature; the model's computation and memory rates are defined independently below.

Chemical atomic clock as a paired coordinate

Let $N(t) = f_{\text{ref}} t$ count reference cycles. Atomic timekeeping supplies an oscillation-based reference. [6] Define the local chemical progress coordinate by

$$\vartheta(z,t) = \int_0^t k_f(T(z,s))\, i(z,s)\,\mathrm{d}s, \qquad \frac{\mathrm{d}\vartheta}{\mathrm{d}N} = \frac{k_f(T) i}{f_{\text{ref}}}. \tag{7.4}$$

With $k_b = 0$, Equation (6.1) integrates exactly to

$$c(z,t) = 1 - [1 - c(z,0)]e^{-\vartheta(z,t)}. \tag{7.5}$$

This expresses the source's chemical-time idea as progress measured against a reference clock.

Logarithmic penetration and inverse-time speed

For a single incident boundary, $k_b = 0$, constant $T, \mu > 0$, $c(z,0) = 0$, and a prescribed attenuated field $I(z) = I_0 e^{-\mu z}$, define a front by $c(z_f,t) = c_\star \in (0,1)$. Wherever the front lies in the interior,

$$z_f(t) = \frac{1}{\mu}\ln\!\left[\frac{k_f I_0 t}{I_\star[-\ln(1-c_\star)]}\right], \qquad \dot z_f(t) = \frac{1}{\mu t}. \tag{7.6}$$

A front exists after the surface reaches its threshold; it reaches the back face at $z_f = H$. The binary exposure readout is $\mathbf{1}_{\{c\ge c_\star\}}$. This formulation makes the logarithmic position law, inverse-time advance, and thresholded state distinct quantities.

8 Clock Alpha and Clock Beta

Coupled fast and slow states

A minimal dual-clock realization uses dimensionless states a, b and input u:

$$\tau_\alpha \dot{a} = -a + \chi b + u, \qquad \tau_\beta \dot{b} = a - b, \qquad 0 \le \chi < 1, \quad 0 < \tau_\alpha \ll \tau_\beta. \tag{8.1}$$

Clock Alpha expresses immediate activity. Clock Beta integrates the active state and feeds retained state back into the next response. Both evolve on the same reference time axis; their different time constants create the dual-clock architecture.

Stable coupled execution. The system matrix is

$$\mathbf{A} = \begin{bmatrix} -1/\tau_\alpha & \chi/\tau_\alpha \\ 1/\tau_\beta & -1/\tau_\beta \end{bmatrix}. \tag{8.2}$$

Its characteristic polynomial is

$$\lambda^2 + \left(\frac{1}{\tau_\alpha} + \frac{1}{\tau_\beta} \right) \lambda + \frac{1 - \chi}{\tau_\alpha \tau_\beta} = 0. \tag{8.3}$$

Both nonleading coefficients are positive, so both roots have negative real part. At constant input u_0, the equilibrium is $a_\star = b_\star = u_0/(1 - \chi)$.

Inverse-logarithmic history register

The source's inverse-log memory can be specified separately in discrete time:

$$w_j = \frac{1}{\kappa_h \ln(j + e)}, \qquad M_\beta[n] = \sum_{i=1}^{n} w_{n-i}\, v_i, \qquad \kappa_h > 0. \tag{8.4}$$

Here v_i is a dimensionless encoded event. Define $W_n = \sum_{i=1}^{n} w_{n-i}$ and a normalized readout

$$\bar{M}_\beta[n] = \frac{M_\beta[n]}{W_n}. \tag{8.5}$$

The extensive register M_β and the normalized readout serve different purposes. Since the normalized weights sum to one, $\left\| \bar{M}_\beta[n] \right\| \le \max_{i \le n} \|v_i\|$. A constant input sequence is reproduced exactly by the normalized readout.

Persistent history

Retention of the original events is specified by the ordered register $\mathscr{H}_n = (v_1, \dots, v_n)$. The slow filtered state is a readout of history; the committed record stores the history itself. These registers give the source's slow interior and frozen core two precise, complementary roles.

Source correspondence: *trewel.pdf*, pp. 26–30..

9 Walnut crosshatch and orthogonal mechanics

Grammar builds the network; the network carries the dynamics

Let an L-system be a symbolic grammar $\mathscr{L} = (V, \omega, \mathscr{P})$ and let its material interpretation be

$$G_k = \mathscr{I}\big(\mathscr{P}^k(\omega)\big) = (\mathscr{V}_k, \mathscr{E}_k). \tag{9.1}$$

Each edge has geometry, stiffness, hydraulic conductance, and orientation. The walnut motif designates this crosshatched, structured medium. State evolution is the coupled conversion, deformation, and transport on G_k, rather than the symbol-rewriting operation alone.

Two designated orientations

For an orthogonal pair of local directions, define

$$\mathbf{R}_q = \mathbf{R}_z(q\pi/2), \qquad q \in \{0, 1\}, \qquad \mathbf{R}_1 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{9.2}$$

Then $\mathbf{R}_q^{\mathsf{T}}\mathbf{R}_q = \mathbf{I}$ and $\det \mathbf{R}_q = 1$. A two-well angular energy is

$$U(\varphi) = B\sin^2(2\varphi) - h\cos(2\varphi), \qquad B > 0, \tag{9.3}$$

with a design range $|h| < 2B$ giving local wells at 0 and $\pi/2$. The bias $h = h(c, p, T)$ couples conversion and pressure to orientation. In the ideal two-state reduction, $q = 0$ minimizes the endpoint energies when $h > 0$ and $q = 1$ when $h < 0$; at $h = 0$ the preceding state may be retained.

Center of mass and lever action

For material density ρ_m,

$$\mathbf{r}_{\mathrm{cm}} = \frac{\int_\Omega \rho_m \mathbf{r}\, \mathrm{d}V}{\int_\Omega \rho_m \, \mathrm{d}V}. \tag{9.4}$$

Under a rigid quarter-turn about a specified pivot $\mathbf{r}_p$,

$$\mathbf{r}'_{\mathrm{cm}} = \mathbf{r}_p + \mathbf{R}_1(\mathbf{r}_{\mathrm{cm}} - \mathbf{r}_p), \qquad \boldsymbol{\tau} = (\mathbf{r} - \mathbf{r}_p) \times \mathbf{F}. \tag{9.5}$$

A lever maps local force to torque and motion. For an ideal quasistatic lever with perpendicular arms, $F_{\mathrm{out}}/F_{\mathrm{in}} = \ell_{\mathrm{in}}/\ell_{\mathrm{out}}$; the displacement ratio is reciprocal. Geometry therefore changes mechanical advantage while preserving ideal work balance.

Source correspondence: *trewel.pdf*, pp. 16–23..

10 Stress rotation and hourglass transport

Tensor covariance

A stress tensor and its rotated representation satisfy

$$\boldsymbol{\sigma}' = \mathbf{R}_1 \boldsymbol{\sigma} \mathbf{R}_1^{\mathsf{T}}. \tag{10.1}$$

For $\boldsymbol{\sigma} = \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}$,

$$\boldsymbol{\sigma}' = \begin{bmatrix} d & -b & -e \\ -b & a & c \\ -e & c & f \end{bmatrix}. \tag{10.2}$$

Similarity preserves the trace, determinant, and eigenvalues. Material loading is specified separately by the constitutive relation

$$\boldsymbol{\sigma} = \mathbb{C}_q \,:\, \big[\boldsymbol{\varepsilon}(\mathbf{u}) - \boldsymbol{\varepsilon}^{*}(c,T)\big] - \alpha_B p \mathbf{I}, \qquad \nabla \cdot \boldsymbol{\sigma} + \mathbf{f} = 0, \tag{10.3}$$

where $\boldsymbol{\varepsilon}(\mathbf{u}) = (\nabla \mathbf{u} + \nabla \mathbf{u}^{\mathsf{T}})/2$ is small strain. Eigenstrain $\boldsymbol{\varepsilon}^{*}$ supplies the conversion-induced swelling or contraction.

Compression and directional permeability

Use tension-positive stress and the scalar compressive measure

$$p_c = \max\{0, -\operatorname{tr}(\boldsymbol{\sigma})/3\}, \quad r_p = r_{p0} e^{-\beta_p p_c}, \quad \mathbf{K}(q, p_c) = e^{-2\beta_p p_c} \mathbf{R}_q \mathbf{K}_0 \mathbf{R}_q^{\mathsf{T}}. \tag{10.4}$$

Here β_p has units Pa^{-1} and $\mathbf{K}_0$ is positive definite. The crosshatch changes transport through compression and through its orientation relative to the driving gradient.

Darcy flux and storage

The porous-flow module is [5]

$$\mathbf{j} = -\frac{\mathbf{K}}{\eta_f}(\nabla p - \rho_f \mathbf{g}), \qquad C_s \dot{p} + \alpha_B \nabla \cdot \dot{\mathbf{u}} + \nabla \cdot \mathbf{j} = s_v. \tag{10.5}$$

C_s is storage compliance (Pa^{-1}), α_B is dimensionless, and s_v is a volumetric source rate (s^{-1}). The mass transported through an oriented section A_g during a dwell is

$$M_{\mathrm{tr}}(X) = \int_0^X \int_{A_g} \rho_f \mathbf{j} \cdot \mathbf{n} \, \mathrm{d}A \, \mathrm{d}t. \tag{10.6}$$

For $\mathbf{K}_0 = \operatorname{diag}(k_{\parallel}, k_{\perp}, k_z)$ and a fixed x-directed gradient, the quarter-turn exchanges $k_{\parallel}$ and $k_{\perp}$. The transport contrast is their ratio; an isotropic network has no orientation contrast.

11 Dwell, memory, and finite recovery

Dwell as accumulated activity

The hourglass duration is the time spent in the designated active orientation,

$$X(t) = \int_0^t \mathbf{1}_{\{q(s)=1\}}\,\mathrm{d}s. \tag{11.1}$$

A bounded memory state may be generated by filling and emptying a finite capacity,

$$\dot{m} = \frac{J_{\mathrm{in}}}{M_\star}(1-m) - \frac{J_{\mathrm{out}}}{M_\star}m, \qquad m \in [0,1], \quad J_{\mathrm{in}}, J_{\mathrm{out}} \geq 0. \tag{11.2}$$

The fluxes are nonnegative filling and draining mass rates. This occupancy is distinct from signed cumulative throughput, which can count repeated circulation.

The source's logarithmic constitutive reduction

Retain the source's explicit dwell-to-memory function in dimensionally typed form:

$$s(X) = \ln(1+\beta_t X), \qquad g(X) = \frac{\eta_t X}{1+s(X)}, \qquad m(X) = 1 - e^{-g(X)}. \tag{11.3}$$

Here η_t and β_t are temporal coefficients. s is a dimensionless constriction proxy, not a stress in pascals. The recovery relation is

$$\boxed{T_{\mathrm{rec}}(X) = \tau_r \ln\bigl(1+m(X)\bigr)}. \tag{11.4}$$

Equations (11.3)–(11.4) specify a reduced constitutive module; a transport realization supplies the parameters or the measured occupancy entering that module.

Monotonic memory and recovery ceiling. For $\eta_t > 0, \beta_t \geq 0$, the function g increases because

$$g'(X) = \eta_t \frac{1 + \ln(1+\beta_t X) - \beta_t X/(1+\beta_t X)}{[1+\ln(1+\beta_t X)]^2} > 0. \tag{11.5}$$

Consequently m and T_{rec} increase, $m(X) < 1$ at finite X, and

$$\lim_{X\to\infty} m(X) = 1, \qquad \lim_{X\to\infty} T_{\mathrm{rec}}(X) = \tau_r \ln 2. \tag{11.6}$$

A drain law realizing the recovery relation

Starting from occupancy m_X, take $\dot{m} = -(1+m)/\tau_r$ until the reservoir is empty, then $\dot{m} = 0$. The solution is $m(t) = (1+m_X)e^{-t/\tau_r} - 1$ up to its first zero. Its emptying time is exactly $\tau_r \ln(1+m_X)$, establishing Equation (11.4) for this declared drain law.

12 The photographic carbon-copy record

Selected exposure becomes a retained state

For shutter interval $[t_0, t_1]$, define the exposure and its recorded representation by

$$\mathscr{E}(\mathbf{r}) = \int_{t_0}^{t_1} E(\mathbf{r}, t)\,\mathrm{d}t, \qquad D = \mathscr{Q}(\mathscr{R}(\mathscr{E}))\,. \tag{12.1}$$

$\mathscr{R}$ is a declared sensor or material response and $\mathscr{Q}$ is the representation and quantization rule. The copied object is the *selected exposure field*, indexed on the image domain. OTAN2 may supply its spatial address.

The source's flash-frozen carbon-copy and carbonite terminology is assigned to the commitment operation

$$\mathrm{Freeze}_{t_\star}(D) = D_\star, \qquad D_\star(t) = D(t_\star) \quad \text{for all subsequent read operations.} \tag{12.2}$$

An orientation-bearing record is the pair $(\mathbf{R}_q, D_\star)$ with $\mathbf{R}_q \in \mathrm{SO}(3)$. Petrification means that the selected record and its declared orientation are fixed in the committed representation.

Idempotent commitment. Once a record has been committed, applying commitment again without a new write leaves it unchanged:

$$\mathrm{Freeze} \circ \mathrm{Freeze} = \mathrm{Freeze}\,. \tag{12.3}$$

This follows directly from the fixed-record definition.

White Dove: stand by the signal

For an additive record representation, on the pair consisting of a retained record and its incoming write increment, define

$$\Psi_{\mathrm{dove}}(D, \Delta D) = (D, 0). \tag{12.4}$$

For the record update $\mathscr{W}(D, \Delta D) = D + \Delta D$, one has

$$\mathscr{W}(\Psi_{\mathrm{dove}}(D, \Delta D)) = D, \qquad \Psi_{\mathrm{dove}}^2 = \Psi_{\mathrm{dove}}. \tag{12.5}$$

The signal names a projection in the record algebra: the existing record is maintained while no new increment is committed. Thermal variables remain governed by the thermal module, and the representation remains governed by its storage rule.

Material and digital realizations

A material mark and a digital array can each provide a realization of $\mathscr{R}$ and $\mathscr{Q}$. Their shared architectural property is the path from selected event to addressable record. The representation makes the exposure event available to Clock Beta and to the next self-referential update.

Source correspondence: *trewel.pdf*, pp. 8–16, 25–27, 48..

13 Self-reference and the master architecture

The next event reads the preceding record

Let p_n be the next prompt or external event, and let $\mathscr{H}_n$ contain the prior events and outputs. The recursive architecture is

$$\begin{aligned} u_n &= \mathscr{E}_{\text{in}}(p_n, \mathscr{H}_n), \\ \mathsf{x}_{n+1} &= \Phi_{\Delta t}(\mathsf{x}_n, u_n), \\ y_n &= \mathscr{G}(\mathsf{x}_{n+1}, u_n), \\ \mathscr{H}_{n+1} &= \mathscr{H}_n \,\|\, (p_n, y_n), \\ D_{n+1} &= \text{Freeze}\big(\text{Encode}(\mathscr{H}_{n+1}, \mathsf{x}_{n+1})\big). \end{aligned} \tag{13.1}$$

$\|$ denotes ordered append. Self-reference has the precise meaning that an update can depend on the stored results of preceding updates. Clock Alpha governs active response; Clock Beta or the inverse-log register supplies a history-dependent readout.

Master state-transition operator

All modules enter a typed transition map

$$\boxed{\mathscr{U}_{\Delta t} : \mathscr{X} \times \mathscr{U} \longrightarrow \mathscr{X}, \qquad \mathsf{x}_{n+1} = \mathscr{U}_{\Delta t}(\mathsf{x}_n, u_n)}. \tag{13.2}$$

Its interfaces are optical selection, co-indexed coordinate encoding, preparaat conversion, orientation-dependent mechanics, transport, memory, and record commitment. The simultaneous continuum system defines the coupling; a numerical implementation declares its solve order or splitting scheme.

Master thread functional

For normalized lane coordinate $\zeta = z/H$ and dimensionless feature fields, define

$$Z_{\text{thread}}[n] = \int_0^1 \Big[w_\alpha \, \|a(\zeta, n)\|^2 + w_\beta \, \|b(\zeta, n)\|^2 + w_m m(\zeta, n)^2 \Big] \, \mathrm{d}\zeta, \tag{13.3}$$

where $w_\alpha, w_\beta, w_m \geq 0$. This scalar measures the partition of the thread's state across activity and retention. Rotations and record projections remain operators on their corresponding components; the scalar functional and the state-transition operator have distinct types.

Architectural closure Every output has an address, every state change has a specified input or constitutive cause, every retained quantity has a storage rule, and every recovery trajectory starts from an accumulated state. The next event enters the same architecture with the preceding history available.

Source correspondence: *trewel.pdf*, pp. 26–29, 47–48..

14 Boundary data and executable closure

A complete instance

An instance supplies geometry $(L_x, L_y, H, w, \ell, \gamma, \phi, d)$, coordinate scales (ϵ, r_0, M), material constitutive functions, clock constants, a representation rule, and an initial state. The thermal and mechanical realizations specify $h(T, \zeta)$, $\mathbb{C}_q$, $\boldsymbol{\varepsilon}^*(c, T)$, $\mathbf{K}_0$, β_p, and the rate functions.

At $t = 0$, prescribe $T, c, \mathbf{u}, p, q, m, a, b, D, \mathscr{H}$ consistently with the constraints. The optical lane specifies $I_0(t)$ and $I_H(t)$. Thermal faces prescribe temperature or heat flux; mechanical faces prescribe displacement or traction; hydraulic faces prescribe pressure or normal flux. A displacement anchor removes unrestricted rigid-body motion, and a pressure reference is included when only pressure differences determine the flow.

A coupled evaluation cycle

1. Apply the slit and tilt-shift operators to the incoming optical field, and evaluate its common coordinate index through OTAN2.
2. Solve the two-direction attenuation equations with the current conversion field; update conversion and thermal state over the selected time step.
3. Update the mechanical constitutive state and the designated orientation; solve force balance and pore-pressure storage with the chosen boundary data.
4. Evaluate Darcy transport, accumulated dwell, and the memory occupancy; update Clock Alpha and Clock Beta from the event representation.
5. Form the output, append the event to history, and commit the resulting representation according to the record rule.

A partitioned implementation repeats the interacting material and mechanical stages until its declared residual tolerance is met. This makes the source's instruction to evaluate both pathways at the same time a coupled update, even when the numerical work is organized into substeps.

Algebraic conformance conditions

The implementation preserves $P_1 + P_2 = P_\star$, keeps $c, m \in [0, 1]$, retains positive permeability and positive time constants, and uses dimensionless arguments for logarithms. It respects quadrant reconstruction, periodic angular indexing, and the origin's angular-validity convention. A rotated tensor retains its invariants; an active constitutive change is represented in its own equation.

Three distinct retained quantities

M_{tr} is signed material throughput; m is finite memory occupancy; $\mathscr{H}$ is an ordered event record. These quantities are coupled by the implementation's encoding, but they are not interchangeable. Their separation is what makes the hourglass, the slow clock, and the frozen history jointly executable.

15 Executed hourglass evaluation

The following values were calculated from Equations (11.3)–(11.4), using the source's numerical choices in normalized time units t_0:

$$\eta_t = 0.45/t_0, \qquad \beta_t = 0.12/t_0, \qquad \tau_r = 5t_0. \tag{15.1}$$

The columns show dwell duration, the dimensionless constriction proxy, memory occupancy, its remaining unfilled fraction, and recovery duration. Keeping the residual visible distinguishes asymptotic filling from a rounded display of 1.0000.

X/t_0	$s(X)$	$m(X)$	$1 - m(X)$	T_{rec}/t_0
1	0.113329	0.33248492	6.675×10^{-1}	1.43522782
12	0.891998	0.94239383	5.761×10^{-2}	3.31960572
23	1.324419	0.98835322	1.165×10^{-2}	3.43653385
34	1.625311	0.99705569	2.944×10^{-3}	3.45836970
45	1.856298	0.99916627	8.337×10^{-4}	3.46365113
56	2.043814	0.99974623	2.538×10^{-4}	3.46510144
67	2.201659	0.99991867	8.133×10^{-5}	3.46553257
78	2.337952	0.99997289	2.711×10^{-5}	3.46566812
89	2.457878	0.99999067	9.330×10^{-6}	3.46571258
100	2.564949	0.99999670	3.296×10^{-6}	3.46572766

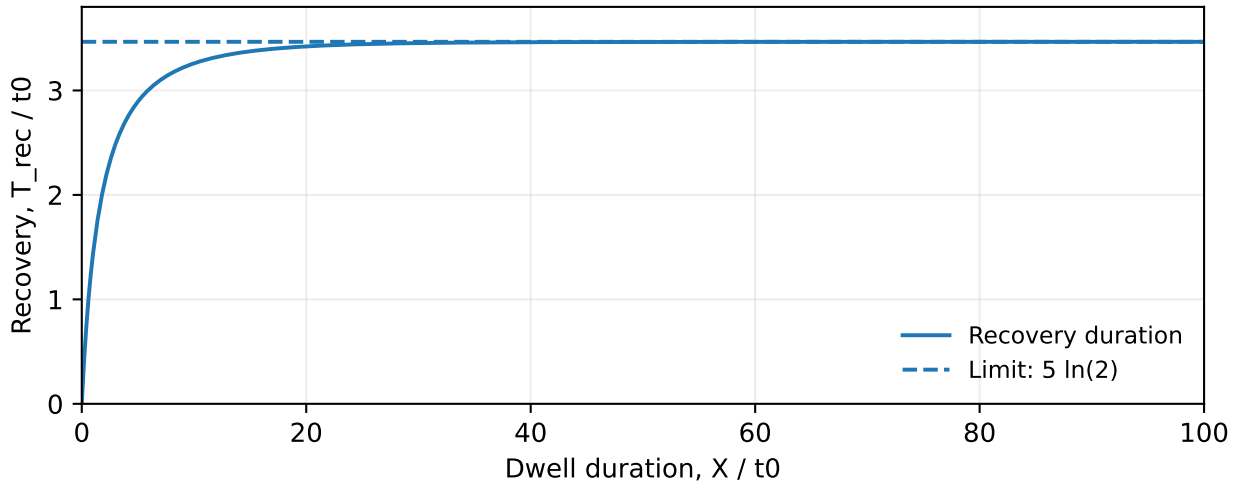


Figure 1. Executed constitutive reduction. The dashed line is the analytical ceiling, not a separate fitted quantity.

$$T_{\mathrm{rec},\infty} = 5 \ln 2 \, t_0 = 3.4657359028 \, t_0. \tag{15.2}$$

At $X = 34t_0$, occupancy is 0.99705569 and the unfilled fraction is approximately 2.944×10^{-3}. At $X = 100t_0$, occupancy is 0.99999670. The reduced architecture approaches its recovery ceiling continuously.

Source correspondence: *trewel.pdf*, the explicit numerical reduction on pp. 29–30; compare its stated interpretation on p. 47..

16 Executed dual-clock and coordinate checks

A declared dual-clock instance

For Equation (8.1), set $\tau_\alpha = t_0$, $\tau_\beta = 20t_0$, $\chi = 0.2$, and $a(0) = b(0) = 0$. The input is $u = 1$ for $0 \le t < 10t_0$ and 0 afterward. Values are evaluated by the matrix exponential on each constant-input interval.

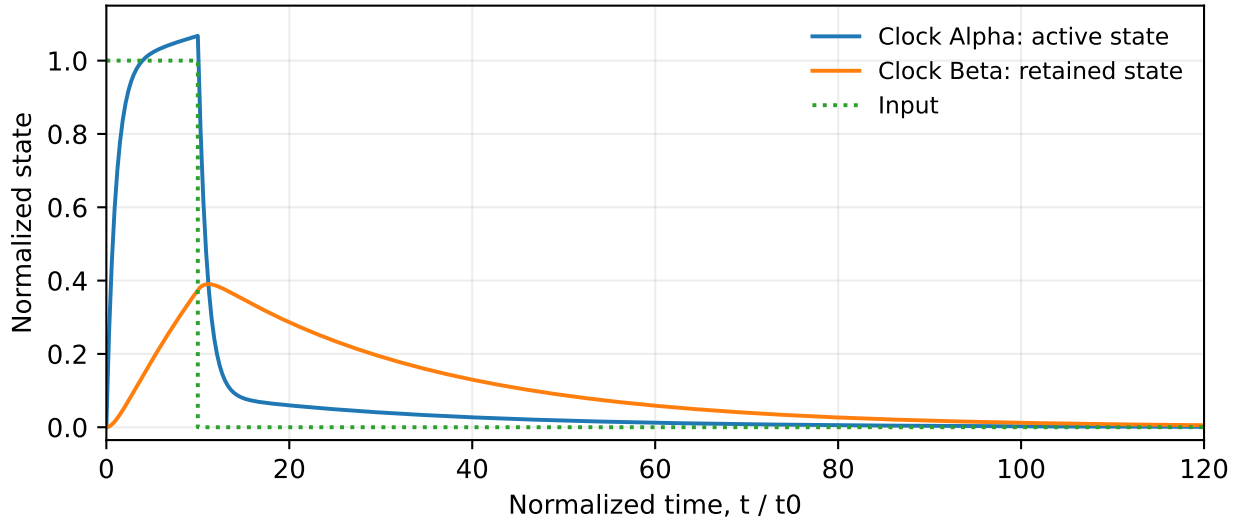


Figure 2. Alpha responds rapidly; Beta rises more slowly and retains activity after the input is released. Feedback permits the active state to exceed the input amplitude.

t/t_0	$a(t)$	$b(t)$
0	0.000000	0.000000
1	0.633143	0.018082
5	1.021470	0.182983
10	1.067607	0.374333
20	0.059653	0.286264
50	0.018179	0.087295
100	0.002511	0.012060

The two decay eigenvalues are

$$\lambda_1 t_0 = -1.01041220, \qquad \lambda_2 t_0 = -0.03958780.$$

They are negative, in agreement with the stability result. At release, $a = 1.06760708$ and $b = 0.37433250$; afterward both relax toward zero on their coupled time scales.

OTAN2 and rotation checks

The reference encoder was evaluated on 20,000 nonzero standard-normal Cartesian pairs using seed 19900710, $\epsilon = 10^{-3}$, $r_0 = M = 1$. The largest sampled wrapped-angle difference from the standard quadrant angle was 4.45×10^{-16} radians or less. The largest normalized radial inverse error was 1.24×10^{-15} or less.

For the numerical symmetric tensor

$$\boldsymbol{\sigma} = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 3 & 0.5 \\ 2 & 0.5 & 5 \end{bmatrix}, \qquad \mathbf{R}_1 \boldsymbol{\sigma} \mathbf{R}_1^{\mathsf{T}} = \begin{bmatrix} 3 & -1 & -0.5 \\ -1 & 4 & 2 \\ -0.5 & 2 & 5 \end{bmatrix}, \tag{16.1}$$

trace agreement was exact in the calculation, and the largest computed eigenvalue difference was 1.78×10^{-15} or less. These checks evaluate the encoder, tensor transform, reduced hourglass law, and linear clock instance specified in this document.

A Reference implementation

The following Python 3.9+ implementation uses the standard library. It evaluates the regularized log-domain encoder and reproduces the hourglass table. Coordinates and scales use a consistent unit, and the output angle is in radians.

```
"""Reference calculations for the Klootwijk architectural specification.
Python 3; only the standard library is required for this module.
"""
from math import atan, exp, expm1, isfinite, log, log1p, pi
from typing import NamedTuple


class PolarRecord(NamedTuple):
    rho: float
    theta: float
    angle_valid: bool


def otan2(y: float, x: float, epsilon: float = 1e-3,
          r0: float = 1.0, scale: float = 1.0) -> PolarRecord:
    """Regularized log-radius and quadrant phase; angles are in radians."""
    if not all(isfinite(v) for v in (x, y, epsilon, r0, scale)):
        raise ValueError("All arguments must be finite.")
    if epsilon <= 0 or r0 <= 0 or scale <= 0:
        raise ValueError("epsilon, r0 and scale must be positive.")
    lr = log(r0)
    lx = log(abs(x)) - lr if x else float("-inf")
    ly = log(abs(y)) - lr if y else float("-inf")
    le = log(epsilon) - lr
    terms = (2 * lx, 2 * ly, 2 * le)
    peak = max(terms)
    rho = 0.5 * scale * (
        peak + log(sum(exp(v - peak) for v in terms)))
    if x == 0 and y == 0:
        return PolarRecord(rho, 0.0, False)
    z = exp(-abs(lx - ly))
    a = atan(z)
    b = pi / 2 - a if ly > lx else a
    c = pi - b if x < 0 else b
    theta = -c if y < 0 else c
    return PolarRecord(rho, theta, True)
```

Reference implementation, continued

```
def hourglass(X: float, eta: float = 0.45,
              beta_t: float = 0.12,
              tau_r: float = 5.0) -> tuple[float, float, float]:
    """Return occupancy, unfilled fraction, and recovery duration."""
    if not all(isfinite(v) for v in (X, eta, beta_t, tau_r)):
        raise ValueError("All arguments must be finite.")
    if X < 0 or eta <= 0 or beta_t < 0 or tau_r <= 0:
        raise ValueError("X, beta_t >= 0; eta, tau_r > 0 required.")
    g = eta * X / (1 + log1p(beta_t * X))
    residual = exp(-g)
    m = -expm1(-g)
    return m, residual, tau_r * log1p(m)


if __name__ == "__main__":
    print("X          m(X)        1-m(X)       T_rec")
    for X in range(1, 101, 11):
        m, residual, recovery = hourglass(float(X))
        print(f"{X:3d}  {m:12.8f}  {residual:12.5e}  "
              f"{recovery:11.8f}")
```

Matrix-exponential clock evaluation

For a constant-input interval, let $x = (a, b)^{\mathsf{T}}$ and $B = (1/\tau_{\alpha}, 0)^{\mathsf{T}}$. The exact interval solution is

$$x(t + \Delta) = x_{\star} + e^{\mathbf{A}\Delta}[x(t) - x_{\star}], \qquad x_{\star} = -\mathbf{A}^{-1}Bu. \tag{A.1}$$

The second numerical figure applies this expression first to $u = 1$ and then to $u = 0$. A numerical matrix exponential evaluates a specified linear system without selecting an integration time step for its dynamics.

Stable numerical forms

`expm1` evaluates $e^{x} - 1$ without cancellation near zero; `log1p` evaluates $\ln(1 + x)$ in the corresponding small-argument regime. The radius uses a log-sum-exp expression so that it does not require squaring the original Cartesian coordinates. The angular reconstruction still distinguishes the origin from a nonzero vector.

B Source concordance

The conceptual source is the author-supplied 49-page exported dialogue, *trewel.pdf*. The map below locates the prompts that introduce the architecture's central connections and the response passages that supply its initial equations. Page numbers refer to the uploaded PDF.

Source pages	Source contribution	Formal location
1–3	Trewel, LNR, OTAN2; OpenCV and DSP	Vocabulary; Sections 4–5
6–7	Two preparaats, opposing surfaces, rectangular field	Sections 3 and 6
8–10	Photograph as carbon copy; absorption and dark record	Sections 6 and 12
11–14	Joint material and coordinate pathways; first coupled equations	Sections 2, 4–7
14–16	Tilt-shift and vertical slit	Section 3
16–18	Walnut lattice, crosshatch lever, quarter-turn, dwell and recovery	Sections 9–11
19–21	Stress rotation and Darcy-style drainage equations	Section 10
22–23	Preparaat activation as the coupled mechanism	Sections 1, 6 and 9
23–25	Temperature and chemical-clock rate	Section 7
25–27	White Dove and retained core	Section 12
26–29	Dual-clock self-reference; Alpha and Beta state equations	Sections 8 and 13
29–30	Explicit numerical hourglass reduction	Sections 11 and 15; Appendix A
47–48	Recovery ceiling and master thread expression	Sections 11, 13 and 15
48–49	Request for a formal report and an actual document	This PDF edition

Edition convention

The source is a dialogue containing both author prompts and assistant elaborations. This edition retains the author's conceptual sequence and assigns consistent notation to its mathematical components. Definitions introduced to complete the architecture—including the typed master state, full quadrant reconstruction, explicit storage roles, and the closed recovery law—are stated in the body of this specification.

Attribution

Tom Klootwijk

NL200678942 / 10-07-1990

The Klootwijk Architectural Paradigm

Version 1.0, 16 October 2026.

References

[S] Tom Klootwijk, author-supplied conceptual dialogue. *trewel.pdf*. 49-page exported conversation supplied with this commission. Source page correspondence is given in Appendix B.

[1] OpenCV. *Geometric Image Transformations*, version 4.13.0 documentation. Definitions of geometric remapping and `warpPolar`. Official documentation. Accessed 16 October 2026.

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[5] COMSOL. *Darcy's Law—Equation Formulation*. Version 6.3 documentation. Definitions of specific discharge, permeability, viscosity, pore pressure, and continuity. Official equation reference. Accessed 16 October 2026.

[6] National Institute of Standards and Technology. *How Do Atomic Clocks Work?* Created 22 August 2024; updated 21 August 2026. Institutional timekeeping reference. Accessed 16 October 2026.

Closing formulation The paradigm is a coupled architecture of selected input, two-preparaat interaction, oriented material response, dual-clock execution, and retained history. Its state is the present configuration together with the record of how that configuration came to be.